BACKUP AND RESTORE PROCESS

SOP-016 | Standard Operating Procedure | Issued under POL-016

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Document Control

Field	Detail
Document Title	Backup and Restore Process
Document ID	SOP-016
Version	1.0 — DRAFT
Parent Policy	POL-016 — Backup Management Policy
Related Documents	POL-002 (Retention), POL-003 (Data Integrity), POL-005 (System Classification), POL-006 (CSV), POL-007 (Change Management), POL-009 (PAM), POL-010 (IS Risk), POL-011 (BC/DR), POL-012 (Incident Response), POL-014 (Audit Trail), SOP-006 (Validation), SOP-007 (Change Management), SOP-009 (PAM), SOP-010 (IS Risk Assessment), SOP-011 (BC/DR Process), SOP-012 (Incident Response), SOP-014 (Audit Trail Review), SOP-015 (ITSM), EXC-001, RACI-001
Owner	CIO
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IMPORTANT: Restore operations on GxP systems are not purely technical events. Every restore on a GxP system has a potential validated state consequence. Compliance / QA must be notified before any GxP restore is executed — without exception. The restore may be legitimate and the validated state may be unaffected, but that determination is Compliance / QA's to make, not Data Custodian (IT)'s.

Revision History

Version	Date	Author	Description
0.1	[DATE]	[Author]	Initial draft
1.0	[DATE]	[Author]	Issued for review

1. Purpose

This SOP defines the operational procedures for backup management and restore execution at [Company Name]. It operationalises POL-016 by establishing the specific steps for backup configuration, daily monitoring and alert response, backup failure management, restore authorisation and execution, restoration testing, GxP-specific backup and restore controls, and the annual backup programme review.

This SOP is the operational partner to SOP-011 (BC/DR Process). SOP-011 governs when and whether to invoke a DR failover. This SOP governs the backup infrastructure and procedures that make DR recovery possible — and that support point-in-time data recovery for scenarios that do not warrant a full DR invocation.

2. Scope

This SOP applies to all backup and restore operations for [Company Name] systems with a tier assignment of Tier 1 through Tier 4, all GxP systems, and all SOX ITGC systems. Backup infrastructure itself (backup servers, vaults, management consoles, encryption key stores) is governed by this SOP as Tier 1 infrastructure. All roles performing actions under this SOP must have completed the relevant training and be listed in the RACI-001.

3. Roles and Responsibilities

Role	Responsibility in This Process
Data Custodian (IT)	Primary operator for all backup and restore activities. Configures and maintains backup schedules. Monitors backup job completion. Responds to backup failure alerts. Executes restore operations on authorisation. Maintains backup infrastructure health. Conducts backup integrity verification. Reports backup programme status to CIO.
CIO	Backup management policy owner. Receives alerts for Tier 1 and GxP backup failures. Authorises Tier 1 and GxP restore operations (with System Owner). Reviews annual backup programme review output. Approves backup infrastructure changes.
Information Security Manager	Co-owner. Confirms backup events are captured in SIEM per SOP-014. Reviews backup access logs. Oversees backup encryption key management. Receives alerts for backup security events.
Compliance / QA	GxP backup and restore oversight. Notified before any GxP restore operation. Confirms validated state assessment post-GxP-restore. Reviews GxP restoration test records as VMF quality records. Co-

Role	Responsibility in This Process
	approves changes to GxP backup configuration.
System Owners	Authorises restore operations for their system (jointly with CIO for Tier 1 / GxP). Confirms data completeness and accuracy post-restore. Notified of backup failures for their systems. Participates in annual backup programme review.
DPO	Notified of backup deletion runs affecting personal data. Confirms no active holds exist before deletion. Advises on GDPR implications of cross-border backup transfers.

4. Backup Configuration and Change Management

Backup configuration is a controlled parameter. Changes to backup configuration — including schedule changes, scope changes, retention policy changes, encryption setting changes, and replication target changes — are Changes under SOP-007 and must follow the appropriate change category.

Step 01	**New System Onboarding — Backup Configuration** When a new system is onboarded or a system's tier is changed, Data Custodian (IT) configures backup in accordance with POL-016 Section 2 before the system enters production: Configuration checklist for each system: — Backup schedule confirmed per system tier — Backup scope confirmed: application data, configuration files, audit trail data (GxP), database transaction logs where applicable — Encryption confirmed: AES-256 at rest, TLS 1.2+ in transit — Offsite / cross-region replication target configured and confirmed active — Air-gap or immutable copy configured (Tier 1 and GxP) — Retention policies configured per POL-016 Section 6 — Backup monitoring alerts configured: failure alerts, missed-job alerts, storage capacity alerts — Backup management system logs forwarded to SIEM (per SOP-014) — First backup job executed and verified manually before system goes live For GxP systems: backup configuration documented in the VMF System Description (SOP-006 Section 3). IQ verification includes confirmation that backup is configured per the specification. OQ test protocol includes a backup and restore test. Compliance / QA co-approves the backup configuration before the GxP system enters production.	Responsible **Data Custodian (IT) / Compliance QA (GxP)**
Step 02	**Backup Configuration Change** Changes to backup configuration for any in-scope system follow SOP-007: — Tier 2–4 backup configuration changes: Standard Change where the change follows the pre-approved configuration template; Normal Change for non-standard modifications — Tier 1 backup configuration changes: Normal Change minimum. CAB review. — GxP backup configuration changes: Normal Change. Validation Impact Assessment mandatory. Compliance / QA co-approval required. If VIA determines impact on validated state: Operational Qualification re-execution or partial validation required before the change is released to production. The pre-change state is always backed up before a backup configuration change is implemented — a misconfigured backup system may result in no backups being taken until the error is detected. The pre-change configuration is documented in the change record.	Responsible **Data Custodian (IT) / Compliance QA (GxP)**

5. Daily Monitoring and Backup Failure Response

Backup monitoring is a continuous and daily operational responsibility — not a weekly task. Data Custodian (IT) reviews backup job status as a start-of-day activity. The monitoring programme operates 24/7 with automated alerting for failures outside business hours.

5.1 Daily Monitoring Routine

Step 01	**Morning Job Status Review** Data Custodian (IT) reviews the backup management console at the start of each business day: — All scheduled backup jobs from the prior 24-hour window: confirmed completed successfully, or flagged as failed / partial — Replication status for Tier 1 and GxP systems: replication lag within threshold (see POL-016 Section 8) — Storage capacity: no storage tier above 80% utilisation — Integrity verification results where overnight verification runs were scheduled Any failed, partial, or missing job is a backup failure event — initiate the failure response process (Section 5.2) immediately. Do not defer investigation to the end of the day.	Responsible **Data Custodian (IT)**

5.2 Backup Failure Response

Step 01	**Failure Detection and Notification** Backup failures are detected via automated alerts (within 15 minutes of failure) or via the daily monitoring review. On detection: Priority classification: — P1 (Critical): failure of any Tier 1 or GxP backup job; replication lag exceeding RPO target for a Tier 1 system; backup storage at 90%+ capacity — P2 (High): failure of any Tier 2 or SOX ITGC backup job; consecutive failures of any Tier 3 backup job (two or more missed runs in sequence) — P3 (Medium): single failure of a Tier 3 backup job; backup integrity verification warning (not failure) — P4 (Low): single failure of a Tier 4 backup job Notification on detection: — P1: Data Custodian (IT) and CIO notified immediately. For GxP systems: Compliance / QA also notified immediately. — P2: Data Custodian (IT) and CIO notified within 30 minutes. — P3/P4: Data Custodian (IT) logs and investigates at next available opportunity within the business day.	Responsible **Data Custodian (IT)**
Step 02	**Failure Investigation** Data Custodian (IT) investigates the root cause of the backup failure: Common failure causes and initial checks: — Backup agent failure: is the backup agent running on the source system? Restart agent and re-run job. Confirm backup completes successfully. — Source system unavailability: was the source system offline during the backup window? Re-schedule the job for the next available window. Assess whether the gap in backup coverage represents an RPO exposure. — Storage capacity: is the backup storage full? Free capacity before re-running. Alert to CIO if capacity increase is required. — Network connectivity: was there a network interruption between the source system and backup storage? Investigate and remediate network fault per SOP-015. Re-run backup once connectivity is confirmed. — Credential failure: did the backup job fail due to an expired service account credential? Rotate credential per	Responsible **Data Custodian (IT)**

	SOP-009 service account management. Re-run backup. — Encryption fault: did the backup encryption configuration error? Do not reduce encryption to work around the fault — investigate and fix the configuration.	
Step 03	**RPO Exposure Assessment** For every P1 or P2 backup failure, Data Custodian (IT) assesses the RPO exposure: — What is the timestamp of the last successful backup for this system? — How long has the system been running without a backup? — Has the gap between the last successful backup and now exceeded the system's RPO target? If RPO is breached or projected to be breached before a new backup can be completed: — CIO notified immediately — System Owner notified: the system is currently operating beyond its recovery point objective — For GxP systems: Compliance / QA notified — an RPO breach on a GxP system means GxP records created since the last successful backup cannot currently be recovered if lost — IS Risk Register entry raised via SOP-010: availability risk, severity based on duration and system tier — SOP-015 problem record raised: root cause investigation to prevent recurrence	Responsible **Data Custodian (IT) / CIO**
Step 04	**Failure Resolution and Verification** On remediation of the root cause: 1. Re-run the failed backup job manually. Confirm successful completion. 2. Confirm the new backup set is complete: scope matches expected scope, audit trail data included (GxP), all expected data volumes present. 3. For P1 failures: Data Custodian (IT) performs a spot-check file or record restore from the new backup set to confirm recoverability — a successful backup job is not the same as a restorable backup. 4. CIO and System Owner notified: backup restored to normal operation, RPO gap closed. 5. Failure event documented in the ITSM system with root cause and resolution. For P1 failures on Tier 1 or GxP systems: problem record raised per SOP-015.	Responsible **Data Custodian (IT)**

6. Restoration Testing

Restoration testing is a mandatory, scheduled programme — not an ad hoc activity. The test schedule is owned by Data Custodian (IT) and reviewed by CIO. Tests are documented as controlled records. GxP restoration tests are quality records filed in the VMF.

Step 01	**Annual Full Restoration Test** Scope: Tier 1 systems. Complete restore to an isolated environment. Procedure: 1. Select the most recent full backup set for the target system. 2. Document the selected backup set: timestamp, backup ID, hash/integrity status. 3. Provision an isolated restore environment: separate from production, no network path back to production during the test. 4. Execute the full restore: application data, configuration, audit trail	Responsible **Data Custodian (IT) / System Owner / Compliance QA (GxP)**

	data. 5. Confirm restore time: record the elapsed time from restore initiation to system-ready. Compare against RTO target — if restore time exceeds RTO, this is a finding. 6. Confirm recovery point: what is the timestamp of the most recent data in the restored environment? Compare against the backup timestamp and the RPO target. 7. Application functional check: confirm the application starts and critical functions operate correctly in the restored environment. 8. Data completeness check: System Owner confirms key records are present and accurate. 9. GxP systems only: Compliance / QA confirms audit trail data is complete, metadata is preserved, and ALCOA+ requirements are satisfied in the restored environment. Test result documented in VMF. 10. Decommission the isolated environment securely after the test — restored data in the test environment is subject to the same access controls as production. 11. Test report: produced by Data Custodian (IT), reviewed by CIO. Findings raised as IS Risk Register entries and problem records.	
Step 02	**Quarterly Partial Restoration Test** Scope: Tier 1 and GxP (quarterly); Tier 2 (biannual). Procedure: 1. Select a random backup set from within the last 30 days — not necessarily the most recent. 2. Select a random sample of records or files from the scope of the backup: minimum 5% of total or 100 records, whichever is smaller. 3. Restore the selected sample to an isolated environment or a staging folder. 4. Confirm each restored item is: complete (all fields / content present), accurate (matches expected values for the recovery point), accessible (can be opened, read, and used). 5. For GxP: confirm audit trail records for the selected sample are included and metadata is preserved. 6. Document results: backup set selected, sample selection method, items tested, pass / fail per item. 7. Findings: any failed item is a backup integrity concern — escalate to CIO and raise an IS Risk Register entry.	Responsible **Data Custodian (IT)**
Step 03	**Monthly Integrity Verification** Scope: Tier 1 and GxP (monthly); Tier 2 (quarterly). Where the backup platform supports cryptographic hash verification: 1. Run the backup platform's integrity verification against the most recent full backup set. 2. Confirm all hashes verify successfully — no segments flagged as corrupted or unreadable. 3. Document the result: backup set, date of verification, result (pass / fail), any segments with failures. 4. Failed hash verification: the backup set is marked as suspect. Do not use as a restore source. Investigate root cause. Attempt to re-create the backup. IS Risk Register entry raised. 5. For GxP systems: DIO reviews monthly integrity verification results. Any failure is a potential ALCOA+ Completeness concern — DIO notifies Compliance / QA.	Responsible **Data Custodian (IT) / DIO (GxP)**

7. Restore Execution

There are three distinct restore scenarios governed by this SOP. Each has a different authorisation chain, a different urgency profile, and a different post-restore obligation. The procedures below address each separately.

7.1 Standard Point-in-Time Restore

A restore of specific data or files for operational reasons — accidental deletion, data corruption, version rollback. Not a DR event. System remains operational throughout.

Step 01	**Restore Request and Authorisation** Any system user may request a restore. The request must include: — What data or files need to be restored — The target recovery point: to what point in time? — The reason for the restore — The destination: restore to original location (overwriting current state) or restore to a staging location for review first? Authorisation: — Tier 2–4 systems: System Owner authorises — Tier 1 systems: System Owner plus CIO authorise — GxP systems: System Owner plus CIO plus Compliance / QA must be notified before execution. Compliance / QA confirms the restore scope will not adversely affect the validated state, or flags validated state implications before the restore proceeds. Authorisation is documented in an ITSM request ticket before the restore is executed.	Responsible **Requester / System Owner / CIO (Tier 1) / Compliance QA (GxP)**
Step 02	**Restore Execution and Verification** Data Custodian (IT) executes the authorised restore: 1. Identify the backup set that contains the target recovery point 2. Confirm backup set integrity (hash verification if available) before using as a restore source 3. For restores to original location: confirm with System Owner that overwriting current data is intended — this is irreversible once executed 4. Execute the restore to the agreed destination 5. Confirm restore completion: data present, complete, and accurate 6. System Owner confirms the restored data is what was required 7. For GxP systems: post-restore, Compliance / QA confirms: (a) audit trail for the restore event is recorded in the GxP system's audit trail, and (b) the restore did not create any audit trail gaps or discontinuities 8. ITSM ticket closed with restore completion timestamp, backup set used (ID and timestamp), and recovery point confirmed	Responsible **Data Custodian (IT) / System Owner / Compliance QA (GxP)**

7.2 Incident Recovery Restore

A restore as part of a security incident recovery (SOP-012) or operational incident recovery (SOP-015). The system may be offline or partially functional. Speed matters — but not at the cost of using a compromised backup.

Step	Action	Owner
1	Identify the appropriate restore point: for security incidents (SOP-012), the restore source must pre-date the compromise. Do not restore from a backup taken during the compromise window — treat all backups within the compromise window as potentially tainted until confirmed otherwise.	Information Security / Data Custodian IT
2	Confirm the selected backup set is clean: integrity hash verified; backup was taken before any indicators of compromise; if ransomware: backup storage was not accessible from the infected system during the infection period.	Data Custodian IT / Information Security
3	Authorise the restore: CIO authorises all incident recovery restores for Tier 1 systems. CISO notified for security incident restores. System Owner informed.	CIO / CISO
4	GxP systems: Compliance / QA notified before restore execution. Validated state assessment will follow restore — per SOP-011 Section 6 Step 4 and Section 7.3 of this SOP.	Compliance QA
5	Execute restore to an isolated environment first where time permits — confirm the restore produces a functional, clean system before cutting over to production.	Data Custodian IT
6	Cut over to restored system. Confirm all critical functions operational.	Data Custodian IT / System Owner
7	Document: backup set used, recovery point, elapsed restore time, any deviations from the restore procedure. All documentation in the SOP-012 / SOP-015 incident record.	Data Custodian IT

7.3 GxP Post-Restore Validated State Assessment

Every restore on a GxP system — whether a standard point-in-time restore, an incident recovery restore, or a DR recovery restore — requires a validated state assessment before GxP activities resume. This is not an additional step Data Custodian (IT) can perform independently — it requires Compliance / QA involvement.

Assessment Step	Action	Owner
1	Technical restore confirmed complete: Data Custodian (IT) confirms the system is running and data is present to the recovery point.	Data Custodian IT
2	Scope of restore assessed: Compliance / QA and Validation Lead review what was restored, from what backup point, and whether any validated configuration parameters were affected.	Compliance QA / Validation Lead
3	Audit trail review: Compliance / QA reviews audit trail records from the restore point to the current time. Confirms the restore event is recorded in the audit trail, and that no audit trail gaps were introduced by the restore.	Compliance QA / DIO per SOP-003
4	Impact assessment: Was the restore triggered by a data integrity event? Were any GxP records overwritten or replaced by the restore? If so, the original and restored versions are both preserved and the change is documented as a deviation per SOP-006 Section 6.3.	Compliance QA Director
5a — Validated state con	Compliance QA Director	

Assessment Step	Action	Owner
firmed intact: Compliance/QA Director issues GxP		

Assessment Step	Action	Owner
Restart Authorisation. GxP activities may resum		

Assessment Step	Action	Owner
e. VMF record created: restore event, scope, re		

Assessment Step	Action	Owner
covery point, authorisation reference.		
5b — Valid	Compliance QA Director / Validation Lead	

Assessment Step	Action	Owner
atedstatenotconfirmed: Systemremainsquara		

Assessment Step	Action	Owner
ntined. NoGxP activities. Deviationraisedper SO		

Assessment Step	Action	Owner
P-006. Remediation plan agreed. Compliance/QA		

Assessment Step	Action	Owner
Director issues GxP Restart Authorisation only a		

Assessment Step	Action	Owner
after remediation is confirmed.		

IMPORTANT: The GxP Restart Authorisation is a quality record. It must be issued by the Compliance / QA Director specifically — it cannot be delegated below that level. Speed of business recovery does not override this requirement.

8. Backup Infrastructure Decommission

When a system is decommissioned, its backup sets do not disappear — they must be managed through the end of the applicable retention period. This is a common programme gap: systems are turned off, and no-one tracks the backup retention obligation.

Step 01	**Pre-Decommission Backup** Before any system is decommissioned: 1. A final full backup is taken and confirmed successful 2. The final backup is labelled as a decommission archive and transferred to long-term archive storage 3. For GxP systems: Compliance / QA confirms the final backup includes complete audit trail data for the full life of the system. Validation Closure Report (per SOP-013 / SOP-006) references the decommission archive backup 4. Retention period confirmed: DPO confirms the retention obligation for the data in the archive. POL-002 schedule applied. 5. Decommission archive registered in the backup retention register with: system name, decommission date, archive location, retention expiry date, DPO sign-off	Responsible **Data Custodian IT / Compliance QA (GxP) / DPO**

Step 02	**Post-Decommission Archive Management** After system decommission: — The decommission archive is retained in read-only storage for the confirmed retention period — Access to the decommission archive is restricted to CIO and Data Custodian (IT). All access logged. — Annual review: Data Custodian (IT) confirms archive storage is intact and accessible. Media viability assessed — long-lived archives on aging media must be migrated before media failure risk becomes material — At retention expiry: DPO confirms no active holds. Secure deletion per POL-002 disposal standards. Deletion event logged.	Responsible **Data Custodian IT / DPO**

9. Annual Backup Programme Review

The annual backup programme review is a holistic assessment of whether the backup programme is delivering the recovery objectives set in POL-011 and POL-016. It is not the same as reviewing backup job success logs — it is a programme-level governance review.

Review Item	Assessment Question	Owner
Coverage completeness	Are all in-scope systems backed up? Have any new Tier 1, Tier 2, GxP, or SOX systems been added since last review that are not yet in the backup programme?	Data Custodian IT
Schedule compliance	Are all systems being backed up at the required frequency? What is the backup job success rate over the past 12 months? How many RPO exposures occurred?	Data Custodian IT / CIO
Restoration test outcomes	Were all scheduled restoration tests completed? What were the results? Were all findings remediated?	Data Custodian IT / CIO
Encryption and security	Is encryption at rest and in transit	Information Security

Review Item	Assessment Question	Owner
	confirmed active across all backup storage? When were encryption keys last rotated? Any backup security events in the period?	Manager
Offsite and air-gap status	Is offsite replication confirmed active for all Tier 1 and GxP systems? Air-gap integrity confirmed quarterly during the year?	Data Custodian IT
Retention compliance	Are backup retention policies configured correctly? Are decommission archives being managed and will expire at the correct time?	Data Custodian IT / DPO
Media and format viability	For long-lived archives (GxP, SOX): are the media and data formats still viable for the remaining retention period? Any migration required?	Data Custodian IT / Compliance QA
Capacity planning	Is backup storage capacity sufficient for the next 12 months at current growth rates?	Data Custodian IT / CIO
GxP programme health	Were all GxP backup-related restoration tests documented in the VMF? Any GxP validated state issues arising from backup or restore events? Any deviations raised?	Compliance QA / DIO
IS Risk Register	Are all known backup programme gaps and residual risks in the IS Risk Register? Are risk treatment plans on track?	Information Security Manager / SOP-010

Annual review report produced by Data Custodian (IT), reviewed by Information Security Manager, approved by CIO. Findings raised as IS Risk Register entries (SOP-010) and change requests (SOP-007) as appropriate. Report retained as a controlled record per POL-002.

10. Related Documents

Document ID	Title	Relationship
POL-016	Backup Management Policy	Parent policy — this SOP operationalises POL-016
POL-011	Business Continuity Policy	RPO targets that the backup programme must

Document ID	Title	Relationship
		deliver. Backup failure to meet RPO is a BC risk assessed under SOP-011.
SOP-011	BC/DR Process	BC/DR invocation uses this SOP's backup programme for restore source. Post-DR restore validated state assessment per Section 7.3 of this SOP.
POL-002	Data Retention and Disposal Policy	Data retention periods define minimum backup archive lifetime. Backup deletion must respect retention holds.
POL-003	Data Integrity Policy	GxP backup completeness and metadata preservation are ALCOA+ requirements.
POL-005	System Classification Policy	System tier determines backup schedule, RPO, offsite requirements, and test cadence.
POL-006	Computer System Validation Policy	GxP backup configuration is a validated parameter per SOP-006. GxP restoration tests are VMF quality records.
POL-007	Change Management Policy	All backup configuration changes are Changes per SOP-007. GxP backup changes require VIA.
POL-009	PAM Policy	Vendor access to backup infrastructure is JIT per SOP-009 with mandatory PAM session recording.
POL-010	IS Risk Policy	Backup programme gaps, RPO exposures, and failed restoration tests are IS risks per SOP-010.
POL-012	Incident Response Policy	Incident recovery restores use pre-compromise backup sets. Incident preservation backups placed under legal hold per CISO instruction.
POL-014	Audit Trail Policy	Backup management system logs forwarded to SIEM. Backup events monitored per SOP-014.
SOP-003	Data Integrity Process	GxP post-restore audit trail review is a DIRA process step per SOP-003.
SOP-006	Computer System Validation Process	GxP backup configuration documented in VMF. Restoration tests are validation activities. GxP restart authorisation per SOP-006 Section 6.4.
SOP-007	Change Management Process	Backup configuration changes follow SOP-007 change categories.
SOP-012	Security Incident Response Process	Incident recovery restore uses pre-compromise backup. SOP-012 governs the security investigation workstream.
SOP-015	ITSM Process	Backup failures are operational incidents under SOP-015. Problem records raised for P1 backup failures.

Document ID	Title	Relationship
EXC-001	Exception Management Policy and Register	Exceptions to backup requirements per EXC-001, subject to constraints in POL-016 Section 10.
RACI-001	Roles, Responsibilities and Authorities	Accountability for backup management governance.

— End of Document —

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